

α -Keto Acid Dehydrogenase Complexes

XX. A KINETIC STUDY OF THE PYRUVATE DEHYDROGENASE COMPLEX FROM BOVINE KIDNEY*

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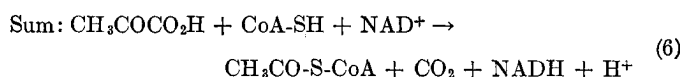
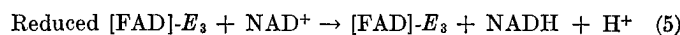
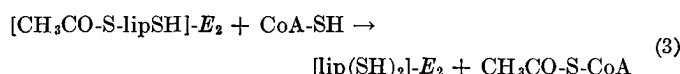
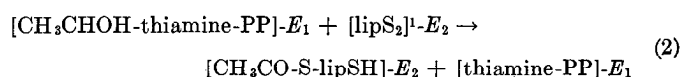
C. STANLEY TSAI,† MICHAEL W. BURGETT, AND LESTER J. REED

From the Clayton Foundation Biochemical Institute and the Department of Chemistry, The University of Texas at Austin, Austin, Texas 78712

SUMMARY

The mammalian pyruvate dehydrogenase complex contains a core, consisting of dihydrolipoyl transacetylase, to which pyruvate dehydrogenase and dihydrolipoyl dehydrogenase (a flavoprotein) are joined. The lipoyl moiety is bound covalently to the transacetylase and, presumably, rotates between the catalytic centers of the three different enzymes that comprise the complex.

The kinetic mechanism of the pyruvate dehydrogenase complex from bovine kidney mitochondria has been investigated. Initial velocity patterns were a series of parallel lines, regardless of which substrate was varied at fixed levels of a second substrate. Product inhibition patterns showed that acetyl-CoA is competitive *versus* CoA and NADH is competitive *versus* NAD, and that both acetyl-CoA and NADH are uncompetitive *versus* pyruvate. These results are consistent with the patterns predicted from rate equations derived by Cleland for three-site ping-pong mechanisms. However, noncompetitive (rather than uncompetitive) inhibition patterns were observed for acetyl-CoA *versus* NAD and for NADH *versus* CoA. Evidence is presented which suggests that these anomalous product inhibition patterns are due to physical association of the flavoprotein with the transacetylase so that combination of acetyl-CoA with the transacetylase hinders combination of NAD with the flavoprotein and combination of NADH with the flavoprotein hinders combination of CoA with the transacetylase.



Pyruvate dehydrogenase (E_1) catalyzes Reactions 1 and 2, dihydrolipoyl transacetylase (E_2) catalyzes Reaction 3, and dihydrolipoyl dehydrogenase (E_3) catalyzes Reactions 4 and 5. E_2 forms the core of the complex, and E_1 and E_3 are attached to it. The lipoyl moiety (lipS_2) is bound covalently to E_2 . We have shown (7) that the lipoyl moiety is attached in amide linkage to the ϵ -amino group of a lysyl residue in the *Escherichia coli* dihydrolipoyl transacetylase. We have suggested (2) that this attachment provides a flexible arm, permitting the lipoyl moiety to rotate between the catalytic centers of the three different enzymes that comprise the pyruvate dehydrogenase complex.

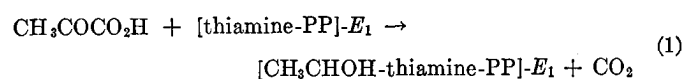
These considerations suggest that over-all Reaction 6 proceeds via a ping-pong mechanism. This paper reports initial velocity and product inhibition patterns obtained with the pyruvate dehydrogenase complex from bovine kidney mitochondria. The results are consistent with the patterns predicted from rate equations derived for three-site ping-pong mechanisms, which are presented in the accompanying paper (8).

EXPERIMENTAL PROCEDURE

Materials—DL-Lipoamide, NAD, NADH, thiamine pyrophosphate, and potassium pyruvate (type III) were purchased from Sigma. CoA and acetyl-CoA were obtained from P-L Biochemicals. Dihydrolipoamide was prepared by reducing lipoamide with sodium borohydride (9). All other chemicals were of the purest commercially available grades. Highly purified preparations of the pyruvate dehydrogenase complex and dihydrolipoyl dehydrogenase were obtained from bovine kidney mitochondria as described previously (10).

Kinetic Studies—The initial rate of over-all Reaction 6 was determined by monitoring NADH formation at 340 nm and 30° with a Gilford recording spectrophotometer. The reaction

The multienzyme pyruvate dehydrogenase complex catalyzes a coordinated sequence of reactions (Reactions 1 to 5) (1-6).

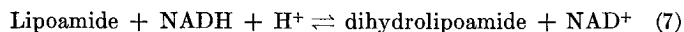


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† On sabbatical from the Department of Chemistry, Carleton University, Ottawa, Canada.

medium contained, in a volume of 1.0 ml, 50 μ moles of potassium phosphate buffer, 0.2 μ mole of thiamine pyrophosphate, 1.0 μ mole of MgCl_2 , 2.6 μ moles of cysteine hydrochloride, and the amounts of pyruvate, CoA, and NAD specified in the text. The pH of the reaction mixture was adjusted to 7.5. After temperature equilibration, the reaction was initiated by the addition of 16 to 20 μ g of enzyme complex. Initial reaction velocities are expressed as micromoles of NADH formed per min.

Initial velocity experiments with dihydrolipoyl dehydrogenase in the complexed (E_1 - E_2 - E_3) and uncomplexed (E_3) forms were carried out by measuring NADH oxidation by lipoamide (Reaction 7). The reaction medium contained, in



a volume of 1.0 ml, 50 μ moles of potassium phosphate buffer, pH 7.5, and the amounts of reactants specified in the text. Lipoamide was added in 0.01 ml of ethanol. After equilibration at 30°, the reaction was initiated by addition of 16 to 20 μ g of enzyme complex or 2.0 to 2.5 μ g of dihydrolipoyl dehydrogenase, and the rate was monitored at 340 nm with a Gilford recording spectrophotometer. Initial reaction velocities are expressed as micromoles of NADH oxidized per min.

In the analysis of the kinetic mechanism, the nomenclature used is that described by Cleland (11, 12).

RESULTS

Initial Velocity Experiments (Reaction 6)—Initial velocity patterns obtained when the concentration of one of the three substrates, pyruvate, CoA, and NAD, was varied at fixed concentrations of the second substrate and at a constant concentration of the third substrate are shown in Fig. 1. The double reciprocal plots gave families of parallel lines, consistent with a ping-pong mechanism for the over-all reaction.

Rate equations for three-site ping-pong mechanisms have been derived by Cleland (8). In the absence of products, these equations reduce to Equation 8, where K_a , K_b , and K_c are Michaelis constants for pyruvate, CoA,

$$v = \frac{V}{1 + \frac{K_a}{[A]} + \frac{K_b}{[B]} + \frac{K_c}{[C]}} \quad (8)$$

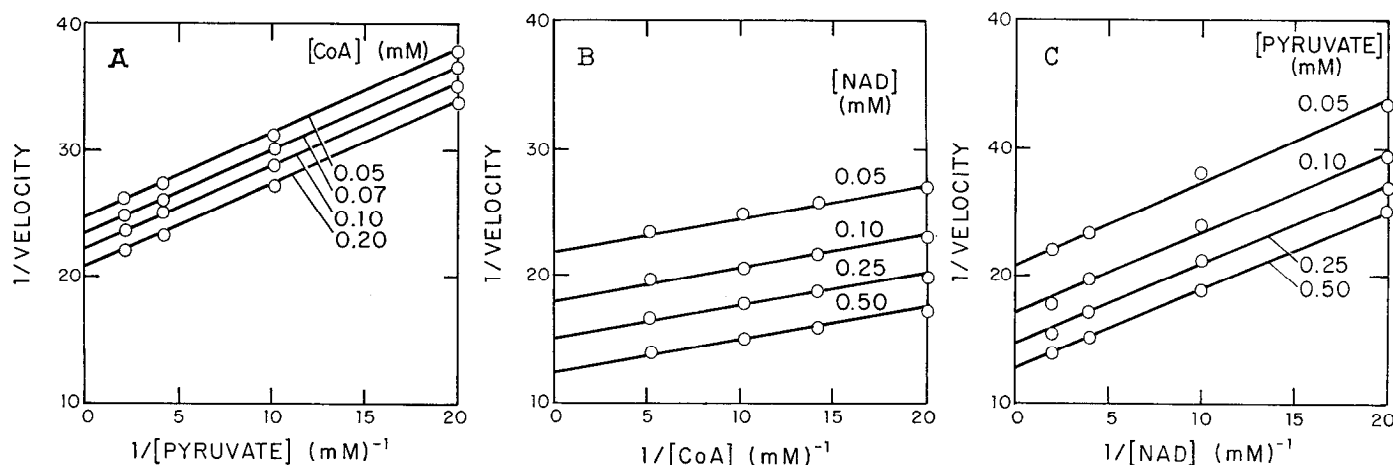


Fig. 1. Initial velocity patterns for the over-all reaction (Reaction 6). A, pyruvate was varied at fixed concentrations of CoA as indicated. NAD was held constant at 0.05 mM. B, CoA was varied at fixed concentrations of NAD as indicated. Pyruvate was held constant at 0.25 mM. C, NAD was varied at fixed con-

centrations of pyruvate as indicated. CoA was held constant at 0.1 mM. Assay conditions were as described under "Experimental Procedure." Velocity is expressed as micromoles of NADH formed per min.

and NAD, respectively. These constants were estimated to be 42 μ M, 20 μ M, and 78 μ M, respectively, from primary plots of reciprocal initial velocities *versus* reciprocal concentrations of pyruvate, secondary plots of primary intercepts *versus* reciprocal concentrations of CoA, and tertiary plots of secondary intercepts *versus* reciprocal concentrations of NAD.

Product Inhibition Experiments (Reaction 6)—The rate equations derived for three-site ping-pong mechanisms (8) predict that each product will be competitive with respect to the substrate that binds at that site and uncompetitive (as long as the individual site reactions are ping-pong; see below) with respect to other substrates. In agreement with these predictions, acetyl-CoA is competitive with respect to CoA (Fig. 2), and NADH is competitive with respect to NAD (Fig. 3). Furthermore, acetyl-CoA and NADH are both uncompetitive with respect to pyruvate (Figs. 4 and 5). However, noncompetitive inhibition patterns were observed for acetyl-CoA *versus* NAD (Fig. 6) and for NADH *versus* CoA (Fig. 7). Similar product inhibition patterns were obtained with preparations of the bovine heart pyruvate dehydrogenase complex (13).

As pointed out by Cleland (8), a product will be noncompetitive against the substrate that binds at the next site in the overall sequence if reaction at this latter site is random sequential. Furthermore, if a product combines as a dead end inhibitor at another site, or in some other way hinders reaction at this other site, it will be noncompetitive *versus* the substrate that normally binds at this site. We have obtained data, presented below, which suggest that the anomalous inhibition patterns observed with acetyl-CoA *versus* NAD (Fig. 6) and with NADH *versus* CoA (Fig. 7) are caused by protein-protein interaction between the transacetylase (E_2) and the flavoprotein (E_3).

Effects of Pyruvate and CoA on Flavoprotein-catalyzed Reaction (Reaction 7)—The pyruvate dehydrogenase complex is organized so that the transacetylase (E_2) forms the core, and the other enzyme components (E_1 and E_3) are attached to it (5). This organization suggested the possibility that protein-protein interactions between E_1 and E_2 and between E_2 and E_3 might affect the reactions catalyzed by the individual enzymes. To investigate this possibility a kinetic study of Reaction 7 was undertaken using the flavoprotein in the uncomplexed state (E_3) and in association with the transacetylase as an integral part of the pyruvate dehydrogenase complex (E_1 - E_2 - E_3). That no inter-

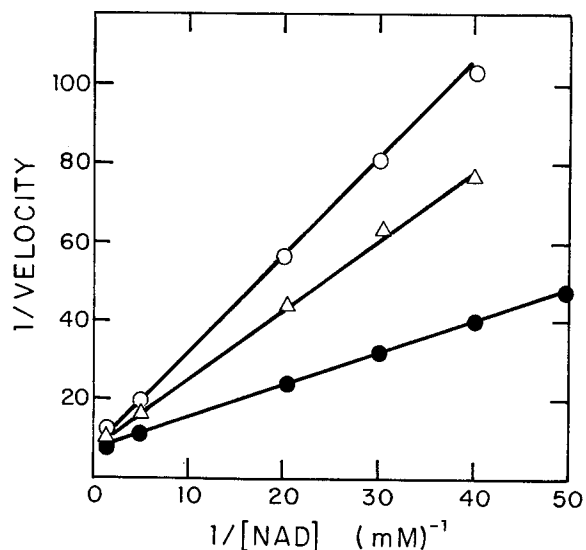
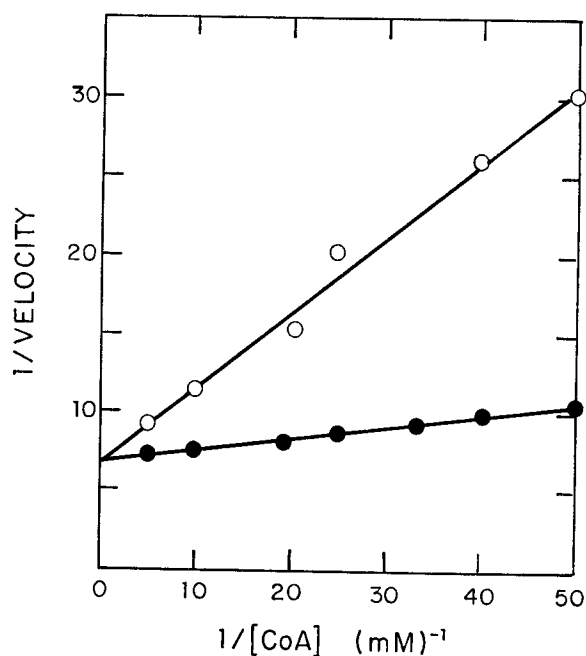


FIG. 2 (top). Product inhibition by acetyl-CoA. CoA was the varied substrate. The concentrations of added acetyl-CoA were 0 (●) and 0.2 mM (○). Pyruvate and NAD were held constant at 2 and 2.5 mM, respectively. Other conditions were as in Fig. 1.

FIG. 3 (bottom). Product inhibition by NADH. NAD was the varied substrate. The concentrations of added NADH were 0 (●), 0.05 (△), and 0.1 mM (○). Pyruvate and CoA were held constant at 2 and 0.13 mM, respectively. Other conditions were as in Fig. 1.

action occurs between E_1 and E_3 in the pyruvate dehydrogenase complex is indicated by the observation that pyruvate (plus thiamine-PP and Mg^{2+}) had no effect on the rate of Reaction 7 (Fig. 8). However, CoA inhibited the complexed flavoprotein (E_1 - E_2 - E_3) noncompetitively with respect to both NADH and lipoamide (Fig. 9). Noncompetitive inhibition patterns were also obtained (data not shown) with CoA versus NAD and with CoA versus dihydrolipoamide (Reaction 7, right to left). Neither CoA nor pyruvate inhibited the uncomplexed flavoprotein (data not shown). These observations suggest that combination of CoA with the transacetylase (E_2) in some way hinders combination of NADH and lipoamide (or NAD and dihydrolipoamide) with the flavoprotein (E_3).

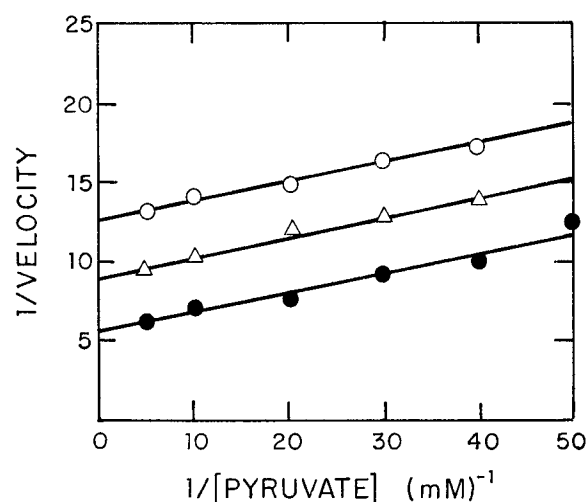
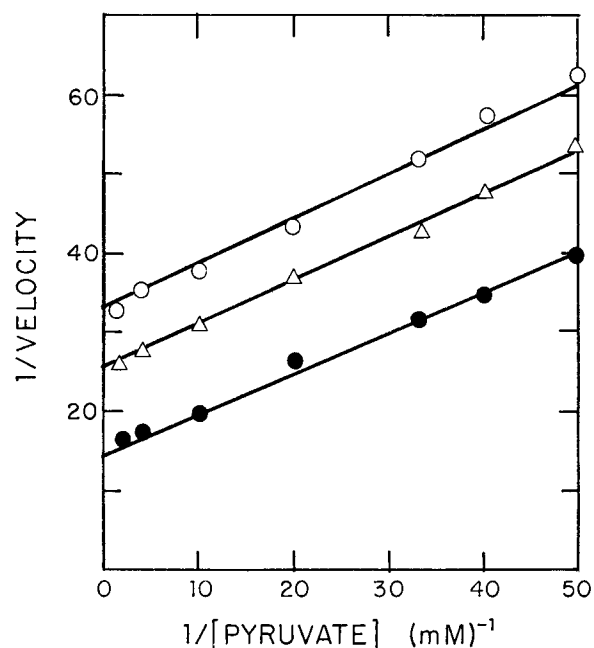


FIG. 4 (top). Product inhibition by acetyl-CoA. Pyruvate was the varied substrate. The concentrations of added acetyl-CoA were 0 (●), 0.2 mM (△), and 0.4 mM (○). CoA and NAD were held constant at 0.13 and 2.5 mM, respectively.

FIG. 5 (bottom). Product inhibition by NADH. Pyruvate was the varied substrate. The concentrations of added NADH were 0 (●), 0.1 (△), and 0.2 mM (○). CoA and NAD were held constant at 0.13 and 1 mM, respectively.

DISCUSSION

Studies on the reactions catalyzed by the pyruvate dehydrogenase complex and its component enzymes have provided good evidence that three different enzymes, acting in sequence, catalyze Reactions 1 to 5 (1-6). These transformations involve interaction of the lipoyl moiety, which is covalently bound to the transacetylase, with α -hydroxyethylthiamine pyrophosphate, which is bound to pyruvate dehydrogenase, and with FAD, which is bound to dihydrolipoyl dehydrogenase. The lipoyl moiety must also interact with CoA at a site on the transacetylase. A possible molecular basis of these interactions was suggested by our finding (7) that the lipoyl moiety is bound in amide linkage to the ϵ -amino group of a lysyl residue in the *E. coli* dihydrolipoyl transacetylase. This attachment provides a flexible arm of about 14 Å for the reactive dithiolane ring, con-

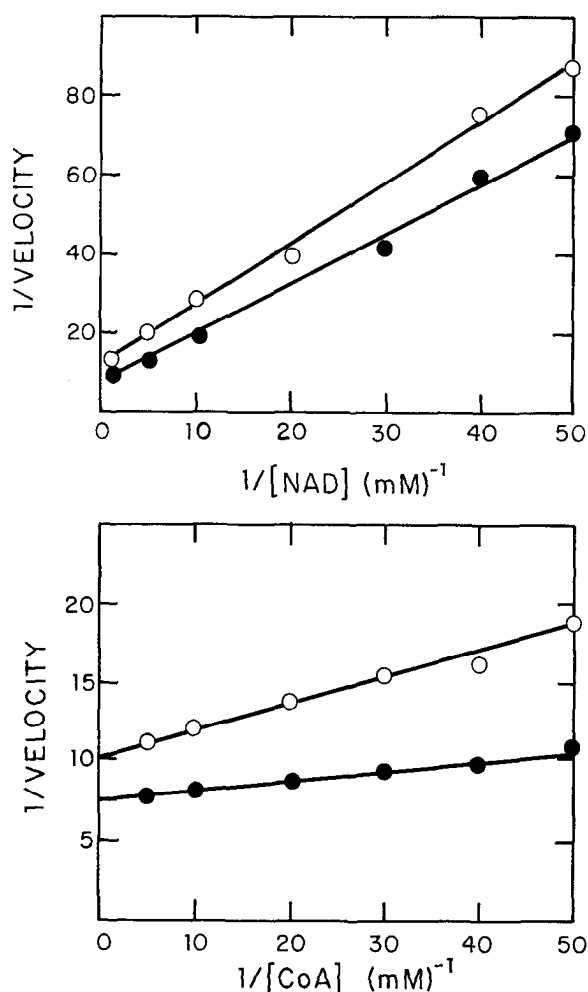


FIG. 6 (top). Product inhibition by acetyl-CoA. NAD was the varied substrate. The concentrations of added acetyl-CoA were 0 (●) and 0.2 mM (○). Pyruvate and CoA were held constant at 2 and 0.13 mM, respectively.

FIG. 7 (bottom). Product inhibition by NADH. CoA was the varied substrate. The concentrations of added NADH were 0 (●) and 0.2 mM (○). Pyruvate and NAD were held constant at 2.0 and 2.5 mM, respectively.

ceivably permitting the lipoyl moiety to rotate between the various catalytic centers (2). These considerations suggest that over-all Reaction 6 proceeds via a three-site ping-pong mechanism.

Although some kinetic properties of the mammalian pyruvate dehydrogenase complex have been reported (14–16), a steady state kinetic analysis of the complex has not been reported prior to the present investigation. During the course of this investigation, Cleland (8) derived rate equations for three-site ping-pong mechanisms. Thus, it has been possible to correlate the observed initial velocity and product inhibition patterns with those predicted by the rate equations. The initial velocity patterns were a series of parallel lines, regardless of which substrate was varied at fixed levels of a second substrate. These results are in agreement with the patterns predicted by the rate equations. The observed product inhibition patterns were also in agreement with those predicted by the rate equations, with two exceptions. Noncompetitive (rather than uncompetitive) inhibition patterns were observed for acetyl-CoA *versus* NAD and for NADH *versus* CoA. A possible explanation of these anomalous results was provided by a study of the flavoprotein-catalyzed oxidation of NADH by lipoamide (Reaction 7). Al-

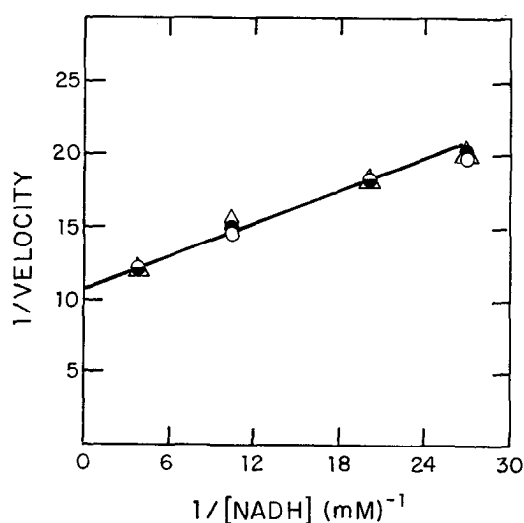


FIG. 8. Effect of pyruvate on flavoprotein-catalyzed reaction (Reaction 7). The oxidation of NADH by lipoamide was measured as described under "Experimental Procedure," with 20 μg of pyruvate dehydrogenase complex, in the absence (Δ) and presence of 0.5 (○) and 1.0 mM (●) pyruvate, 0.2 mM thiamine pyrophosphate, and 0.2 mM MgCl_2 .

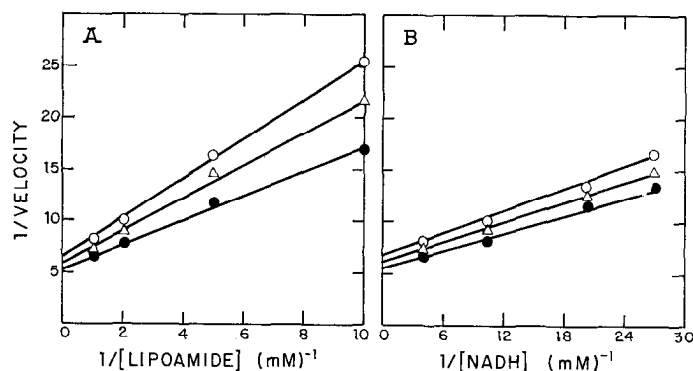


FIG. 9. Effect of CoA on flavoprotein-catalyzed reaction. A, lipoamide was the varied substrate. The concentrations of added CoA were 0 (●), 0.4 (Δ), and 0.8 mM (○). Other conditions were as in Fig. 8. B, NADH was the varied substrate.

though CoA had no effect on the free flavoprotein, it inhibited the flavoprotein in the pyruvate dehydrogenase complex noncompetitively with respect to both NADH and lipoamide. These results suggest that NADH should inhibit the transacetylase in the pyruvate dehydrogenase complex noncompetitively with respect to CoA and that acetyl-CoA should inhibit the complexed flavoprotein noncompetitively with respect to NAD. CoA rather than acetyl-CoA was used in the flavoprotein study to avoid possible complications involving acetylation of the protein-bound dihydrolipoyl moieties and the exogenous dihydrolipoamide catalyzed by the transacetylase (2). There are several possible explanations of these findings. One possibility is that in the pyruvate dehydrogenase complex the catalytic centers of the transacetylase and the flavoprotein are close to each other, so that binding of CoA (or acetyl-CoA) to the transacetylase sterically hinders binding of NADH (or NAD) to the flavoprotein, and vice versa. A second possibility is that binding of CoA (or acetyl-CoA) to the transacetylase causes a conformational change in the transacetylase that hinders binding of NADH (or NAD) to the flavoprotein. Further understanding of these results and their functional significance must

await a more detailed analysis of the structure of the pyruvate dehydrogenase complex (17).

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